

Exercises: Multiple Linear Regression Part I

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(Referring to the theoretical parts: 9, 10, 12, 13, 14, 15, 16, 17, 18, 19)

1 Exercise 1

Among 100 elementary school children, data about daily time spent in front of the TV (TV variable), gender (G variable) and time spent answering to a logic-mathematics question (T variable) were collected.

Exercise 1.1

Specify an appropriate regression model for the response variable T.

$n=100$

$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$

$\uparrow$
Time logic
math questions

$\uparrow$ Time TV

$\uparrow$ GENDER
 $\begin{cases} 1 \text{ MALE} \\ 0 \text{ FEMALE} \end{cases}$

ASSUMPTIONS:

- $E(\epsilon_i) = 0 \quad \forall i=1, \dots, n \rightarrow E[Y_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$
- $\text{VAR}(\epsilon_i) = \sigma^2 \quad \forall i=1, \dots, n$
- $\epsilon_i \sim N(0, \sigma^2)$
- $x_{i1} = x_{i1}; x_{i2} = x_{i2}$
- $\forall \epsilon_i \text{ and } x_{i1}, x_{i2} \text{ are UNCORRELATED} \rightarrow \text{RANK}(X) = p \text{ (Full RANK)}$
- $\text{COV}(\epsilon_i, \epsilon_k) = 0 \text{ for } i \neq k \rightarrow \text{COV}(Y_i, Y_k) = 0$

Exercise 1.2

The model specified at ex 1.1 provides the following values: $SST = 985$, $R^2 = 0.51$, $S.E.(\hat{\beta}_2) = 0.9$, $S.E.(\hat{\beta}_3) = 2.3$, where $\hat{\beta}_2$ and $\hat{\beta}_3$ are the maximum-likelihood estimators of the regression coefficients for TV variable and G variable, and $\hat{\rho}_{(\hat{\beta}_2, \hat{\beta}_3)} = 0.68$.

Perform a statistical test to check the goodness of fit of our model by employing the p-value (specify also the null hypothesis).

$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}$
 ↑
 TIME COEFF
 MEDIA REGRESSIONS
 ↑
 TIME TV
 ↑
 GROSSA
 { 1 male
 0 female

- $SST = 985$
- $R^2 = 0.51$
- $SE(\hat{\beta}_1) = 0.9$
- $SE(\hat{\beta}_2) = 2.3$
- $\rho(\hat{\beta}_2, \hat{\beta}_3) = 0.68$

$\rightarrow SSR = SST \cdot R^2 = 502.35 \rightarrow SSE = SST - SSR = 482.65$

GOODNESS OF FIT

$\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \text{THO} \end{cases}$

$F_{obs} \sim F_{p, n-p}$
 $\uparrow \quad \uparrow$
 $3-1=2 \quad 100-3=97$

$F_{obs} = \frac{\frac{SST - SSE}{p - p_0}}{\frac{SSE}{n - p}} = \frac{\frac{502.35}{3-1}}{\frac{482.65}{100-3}} = 20.479$

$\alpha^{obs} = P(F_{p, n-p} > F_{obs}) \approx 1 - \sim 0$

• SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$
 I REJECT H_0 AT ALL CONFIDENCE LEVEL

• WE DO NOT REJECT THE MODEL

Exercise 1.3

Identify all the elements of the matrix $(X^T X)^{-1}$ which can be computed within the available data (specified in the above exercises).

$$(X^T X)^{-1} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

(3x3)

$$\bullet \text{SE}(\hat{\beta}_1) = 0.9$$

$$\bullet \text{SE}(\hat{\beta}_2) = 2.3$$

$$\bullet \rho(\hat{\beta}_1, \hat{\beta}_2) = 0.68$$

$$\bullet S^2 = \frac{SSE}{n-p} = \frac{482.65}{100-3} = 4.975$$

$$\bullet \rho(\beta_1, \beta_2) = 1 \leftarrow \rho(\beta_1, \beta_1) = 1; \rho(\hat{\beta}_1, \hat{\beta}_2) = 1; \rho(\hat{\beta}_2, \hat{\beta}_2) = 1$$

$$c_{ij} = \rho \frac{\sqrt{\text{VAR}(\hat{\beta}_i)} \sqrt{\text{VAR}(\hat{\beta}_j)}}{S^2}$$

$$\bullet c_{22} = \rho \frac{\sqrt{\text{VAR}(\hat{\beta}_2)} \sqrt{\text{VAR}(\hat{\beta}_2)}}{S^2} = 1 \cdot \frac{0.9 \cdot 0.9}{4.975} = 0.163$$

$$\bullet c_{33} = \rho \frac{\sqrt{\text{VAR}(\hat{\beta}_3)} \sqrt{\text{VAR}(\hat{\beta}_3)}}{S^2} = 1 \cdot \frac{2.3 \cdot 2.3}{4.975} = 1.063$$

$$\bullet c_{23} = \rho \frac{\sqrt{\text{VAR}(\hat{\beta}_2)} \sqrt{\text{VAR}(\hat{\beta}_3)}}{S^2} = 0.68 \cdot \frac{0.9 \cdot 2.3}{4.975} = 0.28$$

$$\bullet c_{32} = c_{23}$$

$$(X^T X)^{-1} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & 0.163 & 0.28 \\ c_{31} & 0.28 & 1.06 \end{bmatrix}$$

(3x3)

2 Exercise 2

Among 100 households in northern Italy, the variables Y = monthly expenditures for foods (in hundreds of euros), X_1 = monthly household income (in hundreds of euros), X_2 = number of household members, and X_3 = type of diet (divided in "vegetarian", "vegan" and "other") were collected.

Exercise 2.1

Specify a multiple linear regression model for the response variable Y .

$N=100$

EXPENDITURE FOODS
INCOME
MEMBERS
1 VEGETARIAN
0 OTHERWISE
1 VEGAN
0 OTHERWISE

$$Y_i = \beta_1 + \beta_2 X_{1i} + \beta_3 X_{2i} + \beta_4 D_{3i} + \beta_5 D_{4i} + \epsilon_i$$

ASSUMPTIONS

- $E(\epsilon_i) = 0 \quad \forall i = 1, \dots, N \rightarrow E[Y_i] = \beta_1 + \beta_2 X_{1i} + \beta_3 X_{2i} + \beta_4 D_{3i} + \beta_5 D_{4i}$
- $\text{VAR}(\epsilon_i) = \sigma^2 > 0$
- $\epsilon_i \sim N(0, \sigma^2)$
- $\text{COV}(\epsilon_i, \epsilon_k) = 0 \quad \forall i \neq k \rightarrow \text{COV}(Y_i, Y_k) = 0 \quad \text{FOR } i \neq k$
- COMPLETENESS $\mathcal{V}_i = X_{1i} \mid \mathcal{V}_{i \neq 1} = X_{2i}$
- THE VECTORS $x_j \in \mathbb{R}^N \quad j=1, \dots, p$ ARE LINEARLY INDEP.
- PARAM. SPACE $\theta = \mathbb{R}^p \cdot \mathbb{R}^T = \mathbb{R}^S \cdot \mathbb{R}^T$
- SAMPLE SPACE $y = \mathbb{R}^N = \mathbb{R}^{100}$

Exercise 2.2

Let $c_{j,h}$ be the elements of the matrix $(X^T X)^{-1}$, where $c_{2,2} = 0.02$, $c_{3,3} = 0.07$, $c_{2,3} = -0.02$. Let also consider that $\hat{\beta}_2 = 0.5$, $\hat{\beta}_3 = 0.8$ and $SSE = 300$.

Evaluate the significance of β_2 and try to interpret the value of $\hat{\beta}_2$.

$$(X^T X)^{-1}$$

$$\bullet c_{2,2} = 0.02$$

$$\bullet c_{3,3} = 0.07$$

$$\bullet c_{2,3} = -0.02$$

$$\bullet \hat{\beta}_2 = 0.5$$

$$\bullet \hat{\beta}_3 = 0.8$$

$$\bullet SSE = 300 \longrightarrow s^2 = \frac{SSE}{n-p} = \frac{300}{100-5} = 3.157$$

$$c_{ij} = \frac{\sqrt{\text{VAR}(\hat{\beta}_i)} \sqrt{\text{VAR}(\hat{\beta}_j)}}{s^2}$$

$$\bullet c_{2,2} = \frac{\sqrt{\text{VAR}(\hat{\beta}_2)} \sqrt{\text{VAR}(\hat{\beta}_2)}}{s^2} \longrightarrow SE(\hat{\beta}_2) = \sqrt{\frac{c_{2,2}}{s^2}} = 0.25$$

$$f(\beta_2, \beta_2) = 1$$

$$\bullet c_{3,3} = \frac{\sqrt{\text{VAR}(\hat{\beta}_3)} \sqrt{\text{VAR}(\hat{\beta}_3)}}{s^2} \longrightarrow SE(\hat{\beta}_3) = \sqrt{\frac{c_{3,3}}{s^2}} = 0.47$$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad t_{obs}^2 \sim t_{n-p}^2 \quad t_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} = \frac{0.5 - 0}{0.25} = 2$$

$$\alpha^{obs} = P(|t_{n-p}| > |t_{obs}|) = 2 \min \{ P(t_{n-p} < -t_{obs}), P(t_{n-p} > t_{obs}) \} \approx 2(1 - \nu) \approx 0.0307$$

• Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$; $\alpha = 0.05$ I REJECT H_0 AT THOSE CONFIDENCE LEVEL

• Since $\alpha^{obs} > \alpha$ for $\alpha = 0.01$ I DO NOT REJECT H_0 AT THIS CONFIDENCE LEVEL

INTERPRETATION OF $\hat{\beta}_2$

• WHEN KEEPING EVERYTHING CONSTANT, ON AVERAGE, IF WE INCREASE THE MONTHLY HOUSEHOLD INCOME BY 1 UNIT (1 THOUSANDS OF EURO) WE INCREASE MONTHLY EXPENDITURES OF FOOD BY 0.5 THOUSANDS OF EUROS

Exercise 2.3

Find the probability distribution of $\hat{\beta}_2 - \hat{\beta}_3$ and build a statistical test based on the null hypothesis $H_0 : \beta_2 = \beta_3$ (at 1% significance level).

$$\hat{\beta}_2 - \hat{\beta}_3 \sim N(\beta_2 - \beta_3; \text{VAR}(\hat{\beta}_2) + \text{VAR}(\hat{\beta}_3) - 2\text{COV}(\hat{\beta}_2, \hat{\beta}_3))$$

$$\alpha = 0.01$$

$$\begin{cases} H_0: \beta_2 - \beta_3 = 0 \\ H_1: 1H_0 \end{cases}$$

$$t_{obs} \stackrel{H_0}{\sim} t_{n-p} \quad t_{obs} = \frac{\hat{\beta}_2 - \hat{\beta}_3 - 0}{\sqrt{\text{VAR}(\hat{\beta}_2) + \text{VAR}(\hat{\beta}_3) - 2\text{COV}(\hat{\beta}_2, \hat{\beta}_3)}} = \frac{0.5 - 0.3}{\sqrt{0.25^2 + 0.47^2 - 2(-0.063)}} = \frac{-0.3}{0.64} = -0.463$$

$$\bullet SE(\hat{\beta}_2) = 0.25$$

$$\bullet SE(\hat{\beta}_3) = 0.47$$

$$\bullet s^2 = \frac{SSE}{n-p} = \frac{300}{100-5} = 3.153$$

$$\text{COV}(\beta_2; \beta_3) = s^2 \cdot c_{23} \rightarrow \text{COV}(\hat{\beta}_2; \hat{\beta}_3) = s^2 \cdot c_{23} = -0.063$$

$$\bullet c_{23} = \frac{\text{COV}(\hat{\beta}_2; \hat{\beta}_3)}{s^2} = \frac{\sqrt{\text{VAR}(\hat{\beta}_2)} \sqrt{\text{VAR}(\hat{\beta}_3)}}{s^2} \rightarrow \sqrt{\text{VAR}(\hat{\beta}_2)} \sqrt{\text{VAR}(\hat{\beta}_3)} = \frac{c_{23} \cdot s^2}{s^2} = -0.02 \cdot 3.153$$

$$\alpha^{obs} = P(|t_{n-p}| > |t_{obs}|) = 2 \min\{P(t_{n-p} < -t_{obs}); P(t_{n-p} > t_{obs})\} \approx 2(1 - \dots)$$

≈ 0.6074 Since $\alpha^{obs} > \alpha$ for $\alpha = 0.01$ I CAN NOT REJECT H_0

Exercise 2.4

Knowing $SSE = 282$ for a model which includes an interaction between the dummy variable (referred to the type "vegetarian") and X_1 , compute and interpret the partial coefficient of determination and decide the best model through an appropriate test (specify hypothesis, test statistic and p-value).

Mod 1:

$$Y_i = \beta_1 + \beta_2 x_1 + \beta_3 x_2 + \beta_4 D_3 + \beta_5 D_4 + \epsilon_i$$

Annotations: EXPENDITURE FOODS (pointing to β_1), INCOME (pointing to β_2), MEMBER (pointing to β_3), {1 VEGETARIAN, 0 OTHERWISE} (pointing to β_4), {1 VEGAN, 0 OTHERWISE} (pointing to β_5)

$$SSE_1 = 300$$

Mod 2

$$Y_i = \beta_1 + \beta_2 x_1 + \beta_3 x_2 + \beta_4 D_3 + \beta_5 D_4 + \beta_6 D_3 x_1 + \epsilon_i$$

Annotations: EXPENDITURE FOODS (pointing to β_1), INCOME (pointing to β_2), MEMBER (pointing to β_3), {1 VEGETARIAN, 0 OTHERWISE} (pointing to β_4), {1 VEGAN, 0 OTHERWISE} (pointing to β_5), {1 VEGETARIAN, 0 OTHERWISE} (pointing to β_6)

$$SSE_2 = 282$$

$$\begin{cases} H_0: \beta_6 = 0 \\ H_1: \beta_6 \neq 0 \end{cases}$$

$$S_{0.05} \overset{H_0}{\sim} F_{p-p_0, n-p}$$

Annotations: $p-p_0 = 6-5$, $n-p = 100-6$

$$F_{obs} = \frac{\frac{SSE_1 - SSE_2}{p - p_0}}{\frac{SSE}{n - p}} = \frac{\frac{300 - 282}{6 - 5}}{\frac{282}{100 - 6}} = 6$$

$$\alpha^{obs} = P(S_{1-p_0; n-p} > S_{obs}) \approx 1 - \alpha = 0.001$$

$$\alpha = 0.05; \alpha = 0.01$$

Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$ we REJECT H_0 at those CONFIDENCE LEVEL $\rightarrow$ PREFER MODEL WITH INTERCEPT

3 Exercise 3

To assess the verbal skills of 33 children, a test was conducted by collecting: the final score, the number of books read monthly by each child, and the number of books read monthly by their parents.

Exercise 3.1

Choose an appropriate response variable together with an appropriate linear regression model. Then, specify the related assumptions and the dimension of the design matrix X .

$n=33$

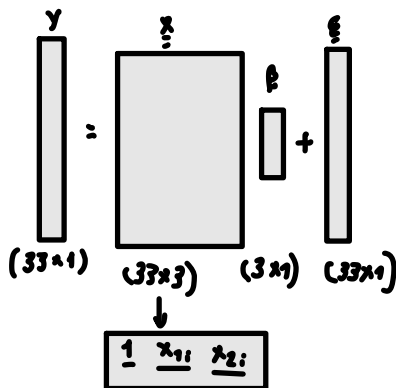
$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$
Final Score # books read child # books read parents

ASSUMPTIONS

- $E(\epsilon_i) = 0 \quad \forall i = 1, \dots, n \rightarrow E[Y_i] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$
- $\text{VAR}(\epsilon_i) = \sigma^2 > 0 \quad \forall i = 1, \dots, n$
- $\text{COV}(\epsilon_i, \epsilon_k) = 0 \quad \text{for } i \neq k \rightarrow \text{COV}(Y_i, Y_k) = 0 \quad \text{for } i \neq k$
- $\epsilon_i \sim N(0, \sigma^2)$
- CONDITIONALLY $\forall i: 1 = x_{1i}; x_{2i} : p = x_{2i}$
- THE VECTORS $x_j \in \mathbb{R}^n$ ARE LINEARLY INDEP. $\rightarrow \text{RANK}(X) = 3$ (FULL RANK)

• PARAM. SPACE: $\Theta = \mathbb{R}^p = \mathbb{R}^4 = \mathbb{R}^3 \cdot \mathbb{R}^1$

• SAMPLE SPACE: $Y = \mathbb{R}^n = \mathbb{R}^{33}$



Exercise 3.2

Complete the following table and provide an interpretation of the estimates of the significant regression coefficients.

	Estimates	S.E.	t-value	p-value
X_1	1.5	0.44	3.409	0.018
X_2	0.605	0.22	2.75	0.01

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad t_{obs}^2 \stackrel{H_0}{\sim} t_{n-p} \\ \uparrow \\ n-p = 30$$

$$t_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{0.44} = \frac{1.5 - 0}{0.44} = 3.409$$

$$\alpha_2^{obs} = P(|t_{n-p}| > |t_{obs}^2|) = 2P(t_{n-p} < -t_{obs}^2) \approx 2(1 - \alpha) \approx 0.018$$

$$\alpha_3^{obs} = 0.01 \rightarrow \text{By knowing } n-p = 30 \text{ AND } \alpha^{obs} = 0.01 \\ \text{Find } t_{n-p} = 2.75$$

$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases} \quad t_{obs}^3 \stackrel{H_0}{\sim} t_{n-p}$$

$$t_{obs}^3 = \frac{\hat{\beta}_3 - \beta_3}{0.22} \rightarrow \hat{\beta}_3 = t_{obs}^3 \cdot 0.22 = 0.605$$

Exercise 3.3

Knowing the SST is equal to 2980 and $R^2 = 0.59$, decide if one of the two below options are compatible with the previous data (considering that the below options are based on a regression model with just one independent variable X_1):

- $SSR = 1800$ and $SSE = 1180$
- $SSR = 1500$ and $SSE = 1500$

Justify your answer.

• $SST = 2980$

• $R^2 = \frac{SSR}{SST} = 0.59 \rightarrow SSR = R^2 \cdot SST = 1758.2 \rightarrow SSE = 1221.8$

FIRST OPTION: $SSR + SSE = SST$ ✓ $\rightarrow$ HOWEVER A MODEL WITH LESS COVARIATES
NEEDS TO HAVE A HIGHER SSE BUT IN THIS
CASE $1180 < 1221.8$ SO NOT SATISFIED

SECOND OPTION:
 $SSR + SSE \neq SST$ ✗

Exercise 3.4

Knowing the SSR is equal to 1440 (for a regression model which just includes X_1), try to evaluate if it is better to include the second independent variable through an appropriate statistical test (specify hypothesis, test statistic and p-value).

Mod 1

$n=33$

$Y_i = \beta_1 + \beta_2 X_{1i} + \beta_3 X_{2i}$

↑ FULL SCORE ↑ BOOM READ CHILL ↑ BOOMS READ PARAGRAPHS

• $SST = 2980$

• $R^2 = \frac{SSR}{SST} = 0.59 \rightarrow SSR = R^2 \cdot SST = 1758.2 \rightarrow SSE = 1221.8$

$n=33$

$Y_i = \beta_1 + \beta_2 X_{1i}$

↑ FULL SCORE ↑ BOOM READ CHILL

Mod 2

• $SST = 2980$

• $SSR_1 = 1440$

• $SSE_1 = 1540$

$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$

$f_{obs}^3 \sim f_{p-p_0; n-p}$

$\uparrow \quad \uparrow$

$3-2=1 \quad 33-3=30$

$f_3^{obs} = \frac{\frac{SSE_2 - SSE_1}{p - p_0}}{\frac{SSE_1}{n - p}} = \frac{\frac{1540 - 1221.8}{3 - 2}}{\frac{1221.8}{33 - 3}} = 7.813$

$\alpha^{obs} = P(f_{p-p_0; n-p} > f_3^{obs}) \approx 1 - \dots \approx 0.009$

Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$
I REJECT H_0 AT ALL CONFIDENCE LEVEL,
I PREFER THE MODEL WITH ALSO X_2

4 Exercise 4

Considering 84 business company in northern Italy, we estimated the following regression model

$$\hat{y} = 12.7 + 9.3x_1 + 1.9x_2 - 1.6x_3$$

where Y = monthly turnover (in thousands), X_1 = sector (1 = manufacturing, 0 = trade), X_2 = number of employees, and X_3 = decrease in investment advertising compared to the previous year (in hundreds of euros). Further, $SSE = 2308$ and $R^2 = 0.62$.

Exercise 4.1

Interpret the estimate of β_2 ($\hat{\beta}_2 = 9.3$).

$n=84$

$$Y_i = 12.7 + 9.3D_i + 1.9x_{2i} - 1.6x_{3i}$$

$\swarrow$ MONTHLY TURNOVER $\swarrow$ SECTOR { 1 MANUFACTURING, 0 TRADE } $\swarrow$ # EMPLOYEES $\swarrow$ DECREASE IN INVESTMENT ADVERTISING

$SSE = 2308$

$R^2 = 0.62$

WHEN KEEPING EVERYTHING CONSTANT, ON AVERAGE,
THE MANUFACTURING SECTOR HAVE A 9.3 THOUSANDS
HIGHER MONTHLY TURNOVER VIRT TRADE

ASSUMPTIONS

$$E(\epsilon_i) = 0 \quad \forall i=1, \dots, n \rightarrow E[Y_i] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

$$VAR(\epsilon_i) = \sigma^2 > 0$$

$$\epsilon_i \sim N(0, \sigma^2)$$

$$COV(\epsilon_i; \epsilon_k) = 0 \quad \text{for } i \neq k \rightarrow COV(Y_i; Y_k) = 0$$

$$\text{CONDITIONALLY } \mathcal{X}_{11} = x_{11}; \mathcal{X}_{1p} = x_{1p}$$

$$\text{VECTORS } x_j \in \mathbb{R}^n \quad j=1, \dots, p \text{ ARE LINEARLY IND } \downarrow$$

$$\text{PARAM SPACE } \Theta = \mathbb{R}^p \cdot \mathbb{R}^1 = \mathbb{R}^1 \cdot \mathbb{R}^p \quad \begin{matrix} \text{RANK}(x) = p \\ \text{(FIX RANK)} \end{matrix}$$

$$\text{SAMPLE SPACE: } Y = \mathbb{R}^n = \mathbb{R}^{84}$$

Exercise 4.2

Complete the table below and show the formula we should use.

	Estimates	S.E.	t-value	p-value
X_3	-1.6	0.6739	-2.3739	0.02

$$\bullet \begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \quad t_{obs} \stackrel{H_0}{\sim} t_{n-p} \quad \alpha^{obs} = 0.02 \rightarrow t_{obs}^3 = -2.3739$$

$df = 1.80$

$$\bullet \hat{\beta}_1 = -1.6 \text{ (GIVEN FROM TEXT)}$$

$$\bullet t_{obs}^3 = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} \rightarrow se(\hat{\beta}_1) = \frac{\hat{\beta}_1}{t_{obs}^3} = 0.6739$$

Exercise 4.3

Evaluate the goodness of fit through a valid test (thus, using the p-value).

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1: \neg H_0 \end{cases}$$

$$f_{obs} \stackrel{H_0}{\sim} F_{p-p_0; n-p} \quad f_{obs} = \frac{\frac{SST - SSE}{p-p_0}}{\frac{SSE}{n-p}} = \frac{\frac{6073.6 - 2308}{4-1}}{\frac{2308}{34-4}} = 43.50$$

$$SSE = 2308$$

$$R^2 = 0.62 = \frac{SSA}{SST} = \frac{SST - SSE}{SST} \rightarrow R^2 SST = SST - SSE \rightarrow SST = \frac{-SSE}{R^2 - 1} = 6073.6$$

$$\alpha^{obs} = P(F_{p-p_0; n-p} > f_{obs}) \approx 1 - 0.000$$

Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ | REJECT $H_0 \rightarrow$ THE MODEL IS GOOD

Exercise 4.4

We would like to assess the added explanatory contribution of the variable "number of employees" compared with the model that does not contain it. Knowing that the regression sum of squares of the model without this variable is equal to 1978, compute an appropriate index/coefficient and interpret the result.

Mod 1.

$$Y_i = 12.7 + 9.3D_1 + 11.9x_2 + 1.6x_3$$

Annotations:
 D_1 : NEWLY TRANSFERRED
 D_1 : SECTOR 1 MANUFACTURING / 0 TRADE
 x_2 : # EMPLOYEES
 x_3 : DECREASE IN INVESTMENTS / INVERTING

$$SSE = 2308$$

$$p = 4$$

Mod 2.

$$Y_i = 12.7 + 9.3D_1 + 1.6x_3$$

Annotations:
 D_1 : NEWLY TRANSFERRED
 D_1 : SECTOR 1 MANUFACTURING / 0 TRADE
 x_3 : DECREASE IN INVESTMENTS / INVERTING

$$SSR = 1978$$

$$SSE = SST - SSR = 6073.6 - 1978 = 4095.6$$

$$p_0 = 3$$

$$SST = 6073.6$$

$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases} \quad f_{obs} \overset{H_0}{\sim} f_{p, p_0; n-p} \quad f_{obs} = \frac{\frac{SSE_2 - SSE_1}{p - p_0}}{\frac{SSE_1}{n - p}} = \frac{\frac{4095.6 - 2308}{4 - 3}}{\frac{2308}{84 - 4}} = 61.96$$

$$\alpha^{obs} = P(f_{p, p_0; n-p} > f_{obs}) \approx 1 - \alpha \approx 0$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I REJECT H_0 . I PREFER THE MODEL W/ # EMPLOYEES WAT MODEL W/OUT # EMPLOYEES